

TBC-ac4C: Identification of N4-acetylcytidine sites by incorporating TCN and BiGRU based on multi-head cross-attention mechanism

Shengli Zhang^{*}, Xinyi Zhang

School of Mathematics and Statistics, Xidian University, Xi'an 710071, PR China

ARTICLE INFO

Keywords:

N4-acetylcytidine sites
Temporal convolutional network
Bidirectional gated recurrent unit
Multi-head cross-attention mechanism

ABSTRACT

N4-acetylcytidine (ac4C), a pivotal post-transcriptional modification, modulates mRNA lifecycle dynamics and is linked to numerous pathologies. Accurate genome-wide identification of ac4C sites is fundamental to functional characterization, but existing in-silico methods require substantial improvement. To address this, we develop TBC-ac4C method. The model leverages a Transformer encoder to capture global dependencies, while parallel temporal convolutional network (TCN) and bidirectional gated recurrent unit (BiGRU) branches extract hierarchical and contextual features, respectively. A multi-head cross-attention mechanism then synergistically fuses these heterogeneous features for a highly discriminative representation. Evaluated on a benchmark dataset derived from acRIP-seq data, TBC-ac4C outperforms the leading state-of-the-art method, STM-ac4C, achieving significant gains of 3.26 % in ACC, 0.0658 in MCC, and 0.0361 in AUC on the independent test set. The model's generalization capabilities are further confirmed on both balanced and imbalanced datasets from MetaAc4C, with particularly strong performance under imbalanced conditions (Sn is 89.29 %, AUC is 0.8962). TBC-ac4C represents a robust and effective tool that will facilitate ac4C-related functional genomics exploration. The datasets and codes are publicly available at <https://github.com/xinyizhang186/TBC-ac4C>.

1. Introduction

RNA chemical modifications constitute a critical layer of post-transcriptional control, with over 170 distinct types across tRNA, rRNA, and mRNA (Boccaletto et al., 2022). Unlike canonical epigenetic marks, they dynamically modulate mRNA stability, transcription, and translation to directly impact cellular functions and disease pathophysiology (Morena et al., 2018; Roundtree et al., 2017; Arzumani et al., 2022; Arango et al., 2018). Among these, N4-acetylcytidine (ac4C) has attracted substantial attention for its high evolutionary conservation, chemical stability, and broad functional relevance (Jin et al., 2020; Thomas et al., 2018). Initially characterized in tRNA and rRNA, ac4C sites have recently been revealed in mRNA by high-throughput sequencing (Ito et al., 2014; Thomas et al., 1978). Catalyzed by acetyltransferase NAT10, it acts as a conserved regulatory hub across species and pathological contexts (Xie et al., 2023; Jiang et al., 2023). Its dysregulation contributes to human pathologies including cancer and metabolic disorders (Bhargava et al., 2018; Zhang et al., 2022). A schematic diagram of ac4C modification site is shown in Fig. 1. In plants, ac4C modulates growth and development via mRNA modification, as

demonstrated in *Arabidopsis thaliana*, tobacco, and maize (Li et al., 2023; Stoeckle et al., 2018; David et al., 2017). Consequently, precise detection of ac4C sites is a fundamental prerequisite for developing novel therapeutics and enhancing crop genetics.

Advances in high-throughput sequencing and machine learning have positioned computational prediction as a robust alternative for ac4C site identification. In 2019, Zhao et al. pioneered this domain with PACES (Zhao et al., 2019) which employed a random forest using position-specific dinucleotide sequence profile (PSDSP) and k-nucleotide frequency (KNF) features. In 2020, Alam et al. proposed XG-ac4C (Alam et al., 2020), integrating trinucleotide pseudo electron-ion interaction potential (PseEIIP) and electron-ion interaction pseudopotentials (EIIP) with XGBoost. In 2021, Wang et al. introduced DeepAc4C (Wang et al., 2021), which utilized the convolutional neural network (CNN) to fuse physicochemical patterns with nucleic acid distribution features. In 2022, Su et al. engineered iRNA-ac4C (Su et al., 2023) to combine k-mer, nucleotide chemical property (NCP), and accumulated nucleotide frequency (ANF) encodings with gradient boosting decision tree (GBDT), outperforming prior models.

The computational identification of ac4C sites has been

^{*} Corresponding author.

E-mail address: shengli0201@163.com (S. Zhang).

<https://doi.org/10.1016/j.compbiolchem.2025.108849>

Received 16 October 2025; Received in revised form 9 December 2025; Accepted 13 December 2025

Available online 14 December 2025

1476-9271/© 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

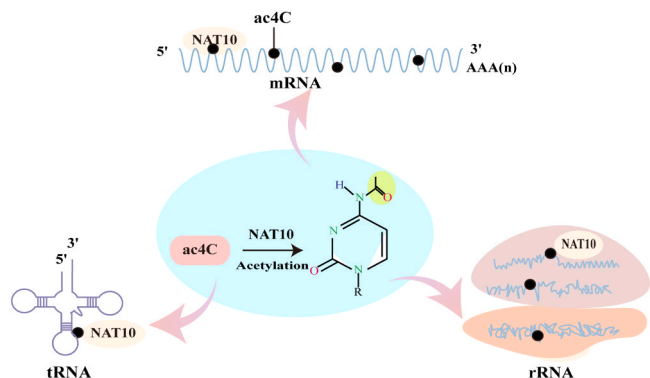


Fig. 1. A schematic diagram of ac4C modification site.

revolutionized by deep learning. In 2023, Lai et al. introduced LSA-ac4C (Lai and Gao, 2023), a framework based on two-layer LSTM and self-attention mechanism, and concurrently developed Auto-ac4C (Lai and Gao, 2023), which leveraged Transformer for automated feature engineering. That same year, Jia et al. build DLC-ac4C (Jia et al., 2023), which integrated DenseNet and BiLSTM. The year 2024 witnessed an explosion of advanced models. ac4C-AFL (Pham et al., 2024) constructed an optimized ensemble predictor from 176 baseline models; DPNN-ac4C (Yuan et al., 2024) adopted a dual-path neural network of CNN and BiGRU; NBCR-ac4C (He et al., 2024) utilized nucleotide Transformer and DNABERT2 for context-aware embeddings, while MetaAc4C (Li et al., 2024) combined the pre-trained BERT with WGAN-GP; STM-ac4C (Yi et al., 2024) focused on multi-scale feature extraction using selective kernel convolution (SKC) and temporal convolutional network (TCN); Voting-ac4C (Jia et al., 2024) integrated the Transformer-based RNAErnie pre-trained large model with six traditional feature encoding methods. This progression continued in 2025

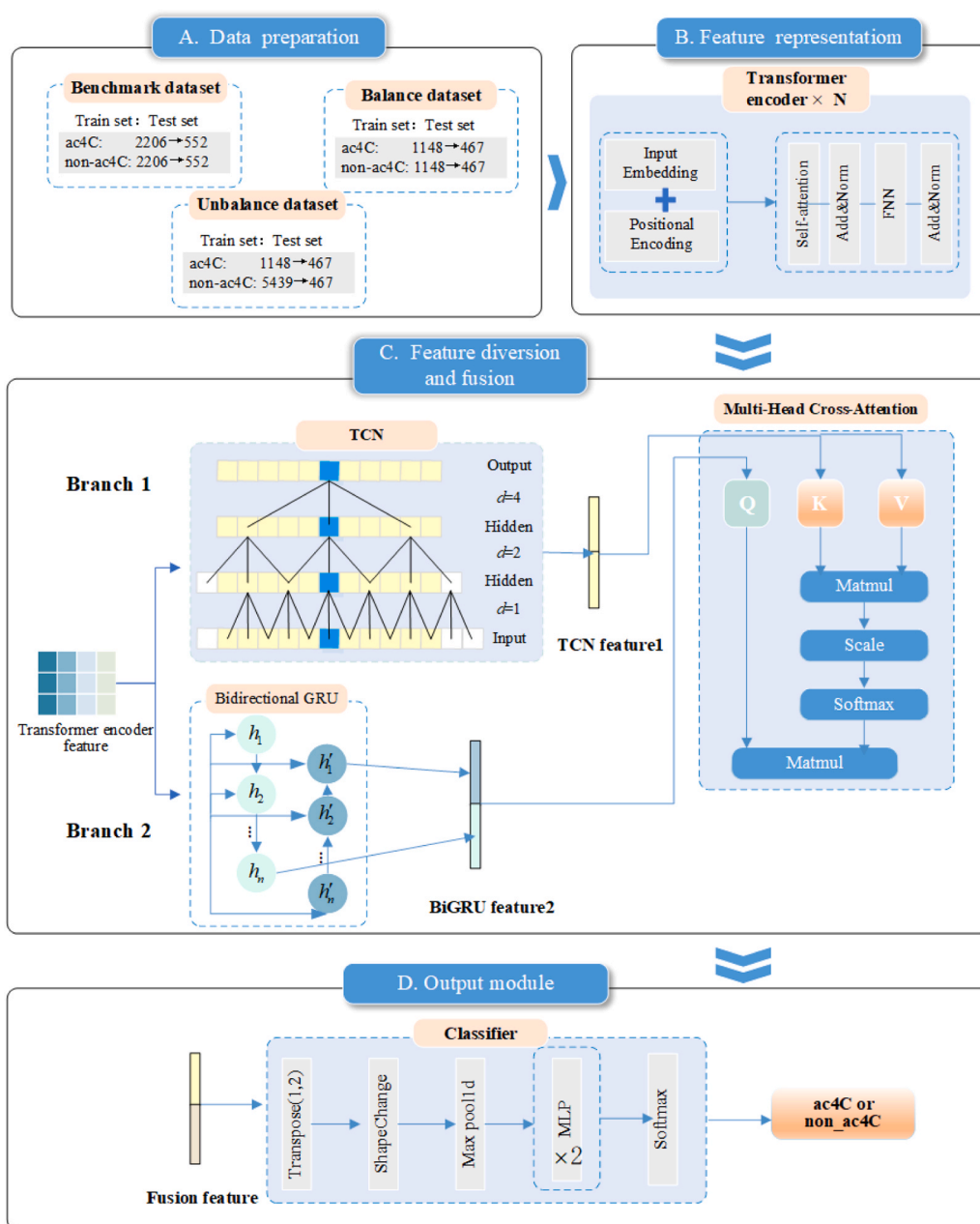


Fig. 2. The flowchart of the TBC-ac4C model.

with Liu et al.'s TransCNN (Zhang et al., 2025), which fused Transformer and TextCNN modules to enhance predictive performance.

Despite significant progress in the field, current deep learning frameworks for ac4C site prediction still face two core limitations. First, most models adopt a single-path architecture (e.g., CNN or RNN), which fails to simultaneously capture both local sequence patterns and long-range dependencies—resulting in either limited local motif recognition or inadequate global context modeling. Second, existing multi-path models typically fuse features using simple concatenation, summation, or self-attention mechanisms. These approaches either neglect inter-branch interactions or focus solely on intra-modality dependencies within a single stream, leading to suboptimal feature fusion and information redundancy.

To address these shortcomings and build upon recent advancements, we have developed TBC-ac4C, an advanced deep learning architecture for superior ac4C site prediction. The novelty of TBC-ac4C lies in its multi-stage, synergistic design, the workflow of which is illustrated in Fig. 2. (i) The architecture first employs a Transformer encoder to capture global, long-range dependencies across the input RNA sequence. This establishes a contextualized foundation for all subsequent feature extraction processes. (ii) A key innovation lies in employing a dual-branch module that concurrently processes Transformer embeddings to learn complementary features. The TCN branch effectively distills hierarchical structures and temporal dynamics, while the BiGRU branch models contextual information bidirectionally in parallel. (iii) To synergize these pathways, we introduce a multi-head cross-attention mechanism. Distinct from simple concatenation that ignores inter-feature relationships and self-attention that only models intra-modality dependencies, multi-head cross-attention enables one branch to query the other, adaptively weighting complementary features. This explicit modeling of interaction between TCN's hierarchical patterns and BiGRU's contextual cues creates a highly discriminative representation, overcoming the limitations of simpler fusion methods. (iv) Critically, comprehensive evaluation on both balanced and imbalanced datasets confirms the model's robustness and generalizability, with performance metrics surpassing existing approaches and overcoming previous methodological constraints.

2. Materials and methods

2.1. Datasets

The current work adopted three datasets to develop and validate our model: a primary benchmark dataset for training and two supplementary datasets for evaluating robustness and generalization. To establish a fair comparative baseline against the STM-ac4C (Yi et al., 2024) model, the evaluation of TBC-ac4C is conducted on the identical datasets. The detailed dataset construction protocol is provided below. The benchmark dataset was constructed based on the work of Su et al. (2023) using acRIP-seq data from human cells published by Arango et al. (2018). Positive samples consisted of 201 bp sequences centered on the cytosine sites within these peaks. Negative samples, of identical length, were generated from randomly selected, non-ac4C modified genomic regions also centered on the cytosine. To reduce homology bias, CD-HIT (Li and Godzik, 2006) was used to remove sequences at a 0.8 identity threshold. The resulting non-redundant dataset contained 2758 sequences (1379 positive and 1379 negative). Subsequently, this dataset was partitioned into the training set (2206 sequences) and the independent test set (552 sequences) at a 8:2 ratio. Furthermore, two additional validation datasets from MetaAc4C (Li et al., 2024) were incorporated, but processed with a stricter CD-HIT (Li and Godzik, 2006) threshold (0.4 identity) to avoid overestimation. The sequences in these datasets featured a length of 415 bp with the ac4C site centrally located. That included a balanced dataset and an imbalanced dataset with a 5:1 negative-to-positive sample ratio, both sharing a common independent test set. Detailed statistics for all datasets are presented in Table 1.

Table 1
Details of the datasets.

Dataset	Train-Test	ac4C	Non-ac4C	Total
Benchmark Dataset	Train	2206	2206	5516
	Test	552	552	
Balance Dataset	Train	1148	1148	3230
	Test	467	467	
Unbalance Dataset	Train	1148	5439	7521

2.2. Transformer-based sequence feature encoding

To numerically represent the input RNA sequences, we first employ an embedding layer. This layer converts each discrete nucleotide (e.g., A, C, G, U) into a continuous, high-dimensional vector representation. The resulting sequence of embedded vectors is then fed into the Transformer encoder block to extract contextualized feature vectors. Adapted from Vaswani et al.'s Transformer model (Vaswani et al., 2013), this approach effectively captures long-range dependencies in biological sequences. The encoder block comprises multi-head self-attention mechanism and feed-forward network, with layer normalization applied to ensure training stability.

The core of the encoder is the multi-head self-attention mechanism, which calculates the relationships between all nucleotide pairs in a sequence. It employs scaled dot-product attention, where Query (Q), Key (K), and Value (V) matrices calculate attention weights, as shown in (1). The attention head outputs are aggregated via concatenation to generate comprehensive features representation in (2).

$$Q_i = XW_i^Q; K_i = XW_i^K; V_i = XW_i^V$$

$$head_i = Self_Attention(Q_i, K_i, V_i) = \text{soft max} \left(\frac{Q_i K_i^T}{\sqrt{d_k}} \right) V_i \quad (1)$$

$$MultiHead(Q, K, V) = Concat(head_1, head_2, \dots, head_h) \quad (2)$$

where W_i^Q , W_i^K and W_i^V are the weight matrices for the i th attention head, and d_k represents the dimension of the key vectors K_i .

To incorporate sequential order, which the Transformer itself does not process, we integrate the positional encoding. This method generates a unique vector for each position using sine and cosine functions in (3). This position encodings are then element-wise fused to the input embeddings.

$$PE_{(pos, 2i)} = \sin \left(\frac{pos}{10000^{2i/d}} \right)$$

$$PE_{(pos, 2i+1)} = \cos \left(\frac{pos}{10000^{2i/d}} \right) \quad (3)$$

where pos represents the position of an element in the sequence, and d is the hidden state dimensionality.

2.3. Model framework

This paper proposes the TBC-ac4C model to address the identification of ac4C sites. Its architecture integrates four core modules, as illustrated in Fig. 2. (1) Feature encoding: a Transformer encoder block converts RNA sequences into contextual embeddings. (2) Parallel feature extraction: the encoded features are processed in parallel by temporal convolutional network (TCN) and bidirectional gated recurrent unit (BiGRU) to temporal dependencies and capture long-range spatial correlations, respectively. This parallel design is to facilitate complementary feature extraction and maintain information fidelity. This approach enables both modules to operate independently on the complete feature space, thereby circumventing the potential "information bottleneck" inherent in serial architectures. The parallel capture of these heterogeneous yet complementary features allow the model to

generate a more discriminative representation for the final fusion stage. (3) Attention fusion: multi-head cross-attention mechanism is employed to adaptively weight and integrate the output representations from the TCN and BiGRU, thereby generating an enhanced feature representation. (4) Dimensionality reduction and classification: the fused high-dimensional features undergo max pooling along the sequence length dimension to enhance and reduce computational overhead. Finally, a two-layer multi-layer perceptron (MLP) is used to output the prediction results.

2.3.1. TCN module

Temporal convolutional network (TCN) (Bai et al., 2018) is incorporated as a parallel branch to extract hierarchical features. In contrast to recurrent neural network (RNN) (Pascanu et al., 2012), TCN is combined with causal convolution, dilated convolution, and residual connection. This design expands the receptive field while preserving temporal causality, thereby effectively capturing long-range dependencies. Given the considerable length of the RNA sequences in this study (201 bp or 415 bp), TCN is particularly well-suited for capturing the interactions between distant nucleotides.

2.3.1.1. Causal convolution. The causal convolution ensures that the output at any position is derived solely from inputs at or before that position. This structure prevents any leakage of information from subsequent positions. To maintain consistent sequence length across layers, zero-padding is applied. The procedure is formally expressed as:

$$y_t = \sum_{i=0}^{k-1} w_i \cdot x_{t-i} + b \quad (4)$$

where w_i is the convolution kernel weight, b is the bias term, and k is the kernel size.

2.3.1.2. Dilated convolution. To efficiently model long-range dependencies, TCN employs dilated convolutions. By stacking layers of causal dilated convolutions, the receptive field of the network grows exponentially with its depth, enabling TCN to process exceptionally long sequences using a comparatively shallow architecture. The mathematical formula is as follows:

$$y'_t = \sum_{i=0}^{k-1} w_i \cdot x_{t-d-i} + b \quad (5)$$

where d is the dilation factor. When $d = 1$, the dilated causal convolution degenerates into a standard causal convolution.

2.3.1.3. Residual block. TCN achieves stable training of deep networks via stacked residual blocks. Every residual unit comprises a pair of dilated causal convolutional layers, with an interspersed activation function, weight normalization, and a dropout layer to enhance non-linear capacity and accelerate convergence. A 1×1 convolution aligns input and output dimensions for residual connections—shortcuts that ensure unobstructed information flow to alleviate the degradation issues prevalent. This design maintains training stability and enhances long-term dependency capture.

2.3.2. BiGRU module

To fully capture the contextual dependencies for each nucleotide, it is essential to process sequence information bidirectionally. We utilize a bidirectional gated recurrent unit (BiGRU) (Li and He, 2019), which extracts both forward and backward contextual features from the time series. Compared to long short-term memory (LSTM) (Pascanu et al., 2012), the GRU possesses a simpler architecture with fewer parameters, which helps to enhance training efficiency. The core operations of the GRU are defined as follows:

$$\begin{aligned} z_t &= \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \\ r_t &= \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \\ \tilde{h}_t &= \tanh(W_h \cdot [r_t \cdot h_{t-1}, x_t] + b_h) \\ h_t &= (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t \end{aligned} \quad (6)$$

where z_t and r_t represent the update and reset gates, respectively. $\tilde{h}_t$ is the candidate hidden state, while h_{t-1} and h_t are the hidden states at the previous and current time steps, respectively. σ represents the sigmoid activation function.

Consequently, the hidden states computed by the forward and backward GRU components are formally defined as:

$$\begin{aligned} \vec{h}_t &= GRU(x_t, \vec{h}_{t-1}) \\ \overleftarrow{h}_t &= GRU(x_t, \overleftarrow{h}_{t+1}) \end{aligned} \quad (7)$$

At time step t , the final output state of the BiGRU hidden layer, h_t is obtained by integrating these two hidden states through an activation function f , as shown below:

$$h_t = f\left(W_{\vec{h}} \cdot \vec{h}_t + W_{\overleftarrow{h}} \cdot \overleftarrow{h}_t + b_t\right) \quad (8)$$

2.3.3. Attention fusion model

In contrast to the self-attention mechanism, which models intra-sequence dependencies, the cross-attention mechanism (Chen et al., 2021) is specifically designed to model the relationships between different sequences. Its core principle is to allow one sequence (referred to as the “query”) to attend to another sequence (the “key” and “value”), thereby enabling the fusion of semantics across sequences.

Cross-attention captures the interactive dependencies between different sequence data by computing a weighted representation of sequence B relative to sequence A . Other processes are consistent with those of the self-attention mechanism. The expression can be formally stated as:

$$\begin{aligned} Q_i &= X_A W_i^Q; K_i = X_B W_i^K; V_i = X_B W_i^V \\ head_i &= Cross_Attention(Q_i, K_i, V_i) = \text{soft max}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i \\ MultiHead(Q, K, V) &= Concat(head_1, head_2, \dots, head_h) \end{aligned} \quad (9)$$

where X_A and X_B represent the input from sequence A and B , respectively.

2.3.4. Output model

The output module progressively transforms latent features X (derived from cross-attention) into class probabilities through two-layer multi-layer perceptron (MLP). The first layer employs a LeakyReLU activation function for nonlinear feature space compression, while the second layer uses a Softmax function to generate the final class probability distribution. The operations are as follows:

$$\begin{aligned} H_1 &= \sigma(W_1 X + b_1), \quad P = \text{Soft max}(W_2 H_1 + b_2) \\ \sigma(x) &= \text{LeakyReLU}(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases} \end{aligned} \quad (10)$$

where W_1, W_2 and b_1, b_2 correspond to the learnable parameters: weights and biases,

and α controls the negative slope.

For the ac4C site identification (binary classification tasks), we optimize the cross-entropy loss with L_2 regularization. The total loss function is formulated as:

$$L = \alpha[-y \log p(y) - (1 - y) \log(1 - p(y))] + \frac{\lambda}{2} \|W\|_2^2 \quad (11)$$

where y denotes the ground truth label, $p(y)$ denotes the model's

predicted probability distribution, and λ is the regularization coefficient.

2.4. Performance metrics

In this paper, we adopt four core binary classification metrics: sensitivity (Sn), specificity (Sp), accuracy (ACC), and the Matthews correlation coefficient (MCC). Sn and Sp quantify the classifier's capacity to correctly discriminate positive and negative samples, respectively. ACC represents the overall classification precision across all samples. MCC measures correlation between true and predicted classifications, offering robust evaluation even for imbalanced datasets. Furthermore, we utilized the receiver operating characteristic (ROC) curve and the precision-recall (PR) curve to visually and quantitatively assess the model's performance. The formulas are defined below:

$$\begin{aligned} Sn &= \frac{TP}{TP + FN} \\ Sp &= \frac{TN}{TN + FP} \\ ACC &= \frac{TP + TN}{TP + TN + FP + FN} \\ MCC &= \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FN) \cdot (TP + FP) \cdot (TN + FP) \cdot (TN + FN)}} \\ AUC \text{ or } auROC &= \int_0^1 TPR(FPR^{-1}(x)) dx \\ auPRC &= \int_0^1 Precision(Recall^{-1}(x)) dx \end{aligned} \quad (12)$$

where TP , FP , TN , and FN refer to true positives, false positives, true negatives, and false negatives, respectively. AUC (auROC) and auPRC are quantified by the area under the ROC curve and the area under the PR curve, respectively.

3. Result and discussion

3.1. Experimental setting and hyperparameter optimization

All models are implemented in Python 3.7 and torch 1.13.1 +cu116, and trained on the High-Performance Computing Platform at Xidian University. We use the Adam (Kingma and Ba, 2014) optimizer for model training. The learning rate is dynamically adjusted according to validation performance via the ReduceLROnPlateau scheduler (PyTorch). Models are trained with 64 for batch size and 100 for the maximum epochs. For a robust and fair evaluation, we utilize the 10-fold cross-validation scheme, consistent with previous studies. This is coupled with an early stopping criterion to prevent overfitting, where training is halted if the validation accuracy does not show an improvement of at least 0.01 % over 15 successive epochs.

We perform automated hyperparameter optimization using the Optuna framework (Akiba et al., 2019), which leverages the Bayesian optimization strategy (Frazier, 2018). The tuning process targeted three key hyperparameters: the embedding dimension, the number of TCN intermediate channels, and the convolutional kernel size, with their search spaces detailed in Table 2. The optimal set of hyperparameters is identified after 30 optimization trials. The optimization results are visualized in Fig. 3. Fig. 3(A) tracks convergence to the optimal parameter set. Fig. 3(B) shows convolutional kernel size exerted the strongest performance impact, followed by embedding dimension, with

TCN channel number least influential. Fig. 3(C) illustrates how hyperparameter values influence model performance to guide optimal range selection. The selected hyperparameters are 64 for embedding dimension, [64, 64, 100] for TCN intermediate channel sizes, and 5 for convolutional kernel size.

3.2. Performance comparison with different existing models

The predictive power of TBC-ac4C is first evaluated by 10-fold cross-validation on the benchmark dataset. The detailed evaluation criteria encompassing ACC, Sn, Sp, and MCC are presented in Table 3, with the corresponding ROC and PR curves shown in Fig. 4. It yields high average metrics: ACC = 84.25 % $\pm$ 3.15 %, Sn = 87.99 % $\pm$ 4.69 %, Sp = 80.51 % $\pm$ 5.77 %, MCC = 0.6898 $\pm$ 0.061, and AUC = 0.9032 $\pm$ 0.0221. Collectively, these results demonstrate the model's robust and stable performance, as evidenced by the low standard deviations across the validation folds.

To further assess its generalization and superiority, we benchmark TBC-ac4C against eleven state-of-the-art methods on the independent test set. These competing predictors included PACES (Zhao et al., 2019), XG-ac4C (Alam et al., 2020), iRNA-ac4C (Su et al., 2023), Auto-ac4C (Lai and Gao, 2023), LSA-ac4C (Lai and Gao, 2023), DLC-ac4C (Jia et al., 2023), STM-ac4C (Yi et al., 2024), DPNN-ac4C (Yuan et al., 2024), ac4C-AFL (Pham et al., 2024), Voting-ac4C (Jia et al., 2024), and TransCNN (Zhang et al., 2025). As summarized in Table 4, TBC-ac4C outperforms all competitors, demonstrating superior results in four key metrics: ACC (88.04 %), Sp (85.87 %), MCC (0.7616), and AUC (0.9440). Notably, its MCC score is 0.0658 higher than the runner-up model named STM-ac4C. While XG-ac4C records the highest Sn at 92.57 %, its low Sp of 59.78 % indicates a high false-positive rate. In contrast, TBC-ac4C demonstrates more optimal balance across all evaluation metrics, underscoring its robust and superior predictive performance. To assess performance differences among the models, we employ the Friedman test and the Wilcoxon signed rank test. The results reveal a statistically significant overall difference in model performance, yielding a p-value of 0.00031. Subsequent Wilcoxon tests further confirm that TBC-ac4C's advantage over the other nine models is also significant, with a p-value of 0.03125 for all pairwise comparisons. The comparison with XG-ac4C yields a p-value of 0.0625, which is significant at the 0.1 level. Collectively, these findings provide strong evidence for the superior overall performance of TBC-ac4C.

3.3. Ablation experiments

To systematically dissect the contribution of each fundamental module to the overall performance of TBC-ac4C, TBC-ac4C model as the baseline, we conduct evaluation of multiple model variants through both standardized benchmarking and independent test set validation: (i) a variant without the TCN module, (ii) a variant without the BiGRU module, (iii) a variant where the cross-attention mechanism is substituted with self-attention, and (iv) a variant without the Transformer encoder, where the initial sequence embeddings are fed directly into the parallel TCN and BiGRU modules. The comparative results are presented in Fig. 5, Table 5 and Table 6. The outcomes clearly indicate that the fully integrated TBC-ac4C achieves the highest performance. Notably, the removal of different components resulted in a graded decline in accuracy, indicating their varying degrees of contribution to the overall model. More specifically, the Transformer encoder module is indispensable. Its removal induces the most substantial performance degradation. on the benchmark dataset, the ACC decreases from 0.8425 to 0.8055, a 4.39 % decrease; on the independent test set, the MCC drops from 0.7616 to 0.6485, and the AUC also declines significantly from 0.9440 to 0.8914. This underscores its foundational role in capturing global contextual information. Secondly, the TCN module emerges as the critical contributor. In the independent test set, its removal causes the most pronounced performance degradation, with the ACC and MCC

Table 2
Hyperparameter search space of TBC-ac4C model.

Hyperparameter	Searched values
Embedding dim	[16,32,64]
TCN middle channels	[(64,64,100), (64,64,128), (128,128,256)]
TCN kernel size	[3,5,7,9]

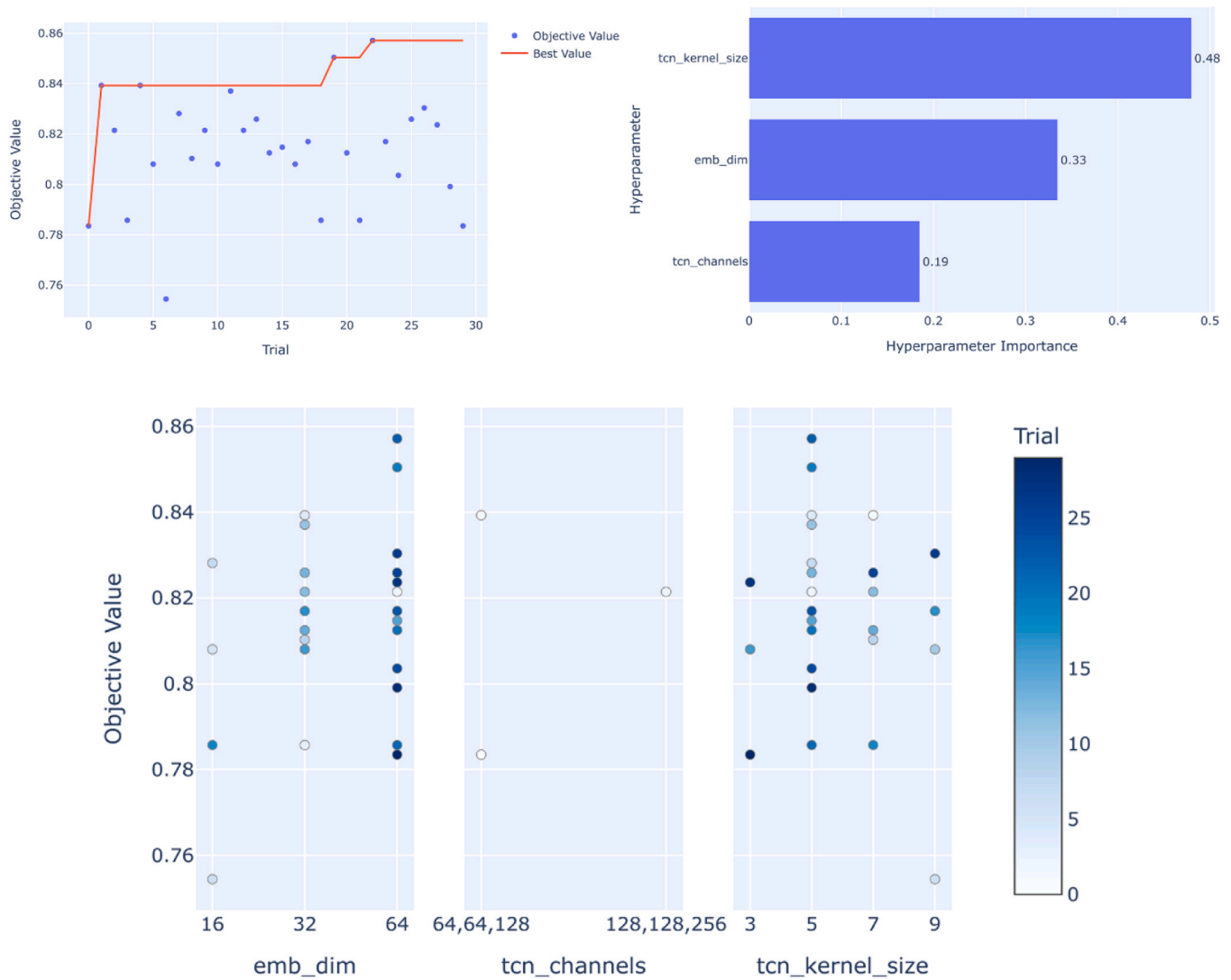


Fig. 3. Hyperparameter optimization results. (A) Optimization history plot. (B) Hyperparameter importance. (C) Quantitative of model performance under different hyperparameter values.

Table 3

The performance of TBC-ac4C in each fold.

10-fold	ACC(%)	Sn(%)	Sp(%)	MCC
1	82.58	87.33	77.83	0.6545
2	78.73	88.24	69.23	0.5853
3	82.31	88.24	76.36	0.6507
4	87.98	94.12	81.82	0.7653
5	85.03	83.71	86.36	0.7010
6	83.45	92.31	74.55	0.6795
7	87.98	89.55	86.43	0.7600
8	86.62	90.00	83.26	0.7342
9	86.85	89.09	84.62	0.7377
10	80.95	77.27	84.62	0.6206
Mean	84.25 ± 3.15	87.99 ± 4.69	80.51 ± 5.77	0.6898 ± 0.061

dropping by 4.16 % and 0.0761, respectively. This underscores the TCN's essential function in effectively capturing long-range sequential dependencies. The impacts of removing the BiGRU layer or substituting the attention mechanism are less pronounced but still significant. This further highlights the synergistic advantage of the parallel architecture. By independently extracting complementary features and adaptively fusing them, the model achieves significantly enhanced predictive performance. Therefore, these results validate our decision to employ the

complete, synergistic architecture as the final TBC-ac4C model.

3.4. Visualization of feature dynamics

To elucidate the feature learning trajectory of the TBC-ac4C model, the high-dimensional feature representations acquired at distinct training stages are visualized. Using uniform manifold approximation and projection (UMAP) (McInnes et al., 2018), we project the features extracted from the initial, intermediate (epoch 10), and final training stages into a 2D space. Fig. 6 demonstrates a clear progression of results. Initially, the feature embeddings of positive and negative samples are highly overlapping and largely indistinguishable. After just 10 epochs, noticeable separation between the two classes begins to form, indicating that the model is learning discriminative patterns. By the end of training, the samples resolve into two distinct and compact clusters with high inter-class separability. This demonstrates that the TBC-ac4C model successfully learns powerful and highly separable features, thereby confirming its effectiveness for this classification task.

To investigate the sequence conservation of ac4C modification sites, we perform motif analysis using the MEME tool (Bailey et al., 2015). As illustrated in Fig. 7, three conserved motifs with highly significant statistical values are identified (E-values of 6.8e-190, 7.0e-67, and 5.4e-27), indicating these motifs are non-randomly distributed and

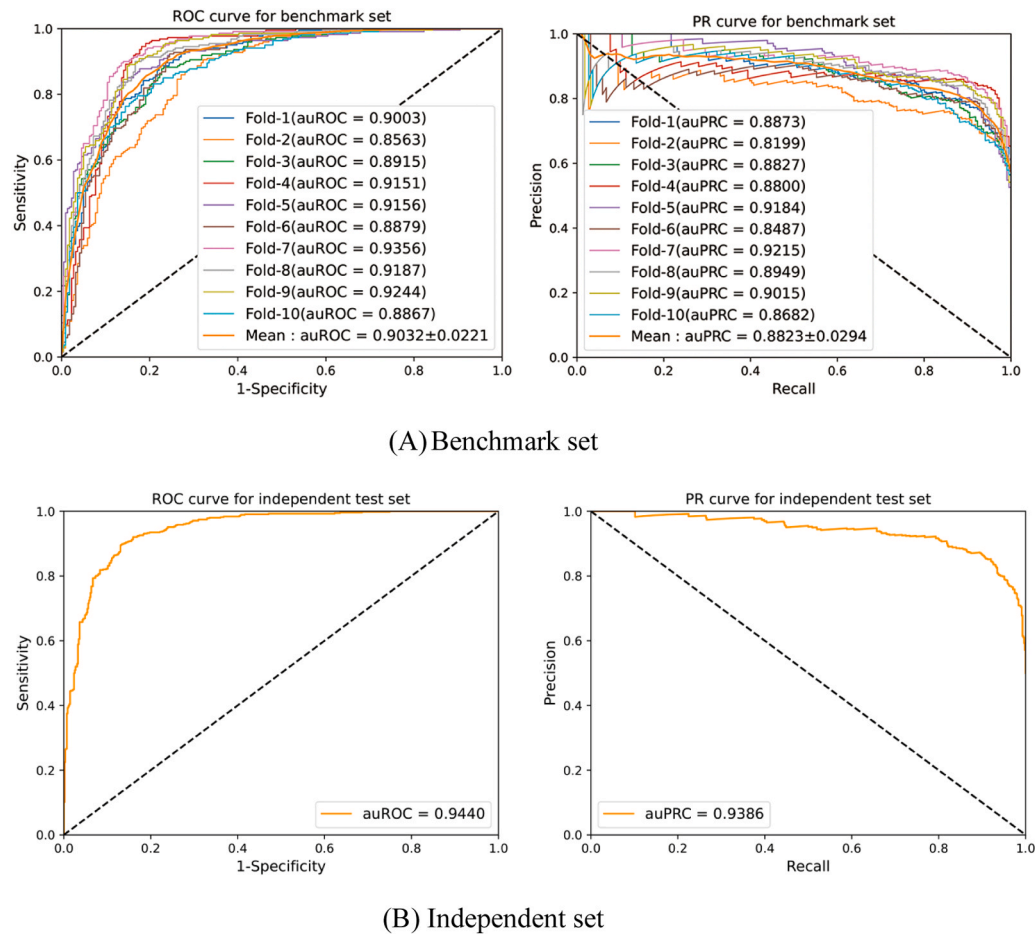


Fig. 4. ROC and PR curves for TBC-ac4C model.

Table 4
Comparative evaluation of multiple methods on the independent test set.

Method	ACC(%)	Sn(%)	Sp(%)	MCC	AUC
PACES	78.80	79.71	77.90	0.5762	0.8648
XG-ac4C	76.18	92.57	59.78	0.5542	0.8713
iRNA-ac4C	79.81	76.70	82.91	0.5970	0.8800
Auto-ac4C	80.71	82.61	78.80	0.6146	0.8894
LSA-ac4C	82.70	87.13	78.26	0.6566	0.8953
DLC-ac4C	82.97	86.23	79.71	0.6608	0.9042
STM-ac4C	84.78	85.87	83.70	0.6958	0.9079
DPNN-ac4	82.78	81.78	84.78	0.6578	0.9103
ac4C-AFL	82.30	84.40	80.30	0.6470	0.8950
Voting-ac4C	83.15	85.14	81.15	0.6635	0.8873
TransCNN	82.89	-	-	0.6597	0.8925
TBC-ac4C	88.04	90.22	85.87	0.7616	0.9440

The bold font denotes the optimal value obtained in each column.

likely possess biological functions. Specifically, Motif 1 presents a G-rich feature with a distinct “GXX” pattern, consistent with the guanine-enriched sequence features revealed by KpLogo analysis in the STM-ac4C (Yi et al., 2024). Motif 2 displays a “CXX” pattern that aligns with the motifs reported in iRNA-ac4C (Su et al., 2023) and DPNN-ac4C (Yuan et al., 2024). Motif 3 demonstrates a hybrid pattern of G and C, showing high conservation at approximately the 20th nucleotide position. Collectively, the identification of motifs provides robust evidence for elucidating the sequence determinants of ac4C modification.

3.5. Robustness and generalization analysis on balanced and imbalanced datasets

To evaluate the robustness and generalization of TBC-ac4C, we train and test the model on the balanced and imbalanced datasets from MetaAc4C (Li et al., 2024), using a common independent test set for evaluation. The model’s performance is benchmarked relative to reference under identical conditions, with detailed comparisons presented in Table 7 and Table 8. On the balanced dataset, TBC-ac4C exhibited superior performance, outperforming all competing models across four key metrics: 84.48 % for ACC, 83.30 % for Sn, 0.6897 for MCC, and 0.9080 for AUC. Especially, TBC-ac4C surpasses the runner-up model named STM-ac4C (Yi et al., 2024), with 2.47 % higher ACC, 0.86 % higher Sn, and 4.07 % higher Sp, affirming its robustness on uniformly distributed data. Similarly, the Friedman test reveals a significant overall performance difference among the models, with a p-value of 0.02163. Post-hoc pairwise Wilcoxon tests yield p-values of 0.0625, 0.09375, 0.03125, 0.03125, 0.03125, and 0.03125 when comparing TBC-ac4C to the other models. All these results are significant at the 0.1 level, indicating that TBC-ac4C maintains a robust and superior predictive capability on the balanced dataset.

On the more challenging imbalanced dataset, TBC-ac4C excels in identifying positive instances, achieving the highest Sn (89.29 %) and AUC (89.6 %). While its ACC (82.98 %) is marginally lower than that of STM-ac4C (83.19 %), this reflects a trade-off where high Sn is achieved at the expense of Sp (76.66 %). This observation also highlights a potential avenue for future model refinement. Collectively, these results validate that the TBC-ac4C model maintains high predictive performance and generalizes effectively across varying data distributions. The



Fig. 5. Results of the ablation experiment for TBC-ac4C.

Table 5
Results of the ablation experiment for TBC-ac4C on the benchmark dataset.

Model	ACC (%)	Sn (%)	Sp (%)	MCC	AUC
Without Transformer Encoder	80.55	86.81	74.30	0.6167	0.8712
Without TCN	82.21	85.63	78.78	0.6474	0.8852
Without BiGRU	83.21	85.40	81.00	0.6663	0.8907
With Self-Attention fusion	83.28	88.25	78.48	0.6707	0.8943
TBC_ac4C	84.25	87.98	80.51	0.6898	0.9032

The bold font denotes the optimal value obtained in each column.

Table 6
Results of the ablation experiment for TBC-ac4C on the independent test set.

Model	ACC (%)	Sn (%)	Sp (%)	MCC	AUC
Without Transformer Encoder	82.25	87.50	76.99	0.6485	0.8914
Without TCN	83.88	91.49	76.27	0.6855	0.8988
Without BiGRU	84.87	89.31	80.43	0.7002	0.9187
With Self-Attention fusion	86.32	88.41	84.24	0.7271	0.9211
TBC_ac4C	88.04	90.22	85.87	0.7616	0.9440

The bold font denotes the optimal value obtained in each column.

Friedman test indicates a significant overall performance difference at the 0.1 level, with a p-value of 0.09197. The corresponding Wilcoxon p-values for TBC-ac4C against other models are 0.09375, 0.09375, 0.15625, 0.15625, 0.1875, and 0.6875. These findings suggest that TBC-ac4C maintains comparable overall competitiveness with state-of-the-art methods while leading in Sn and AUC, making it critically valuable for biological applications where minimizing false negatives is a

priority.

4. Discussion

This section synthesizes the above experimental findings and delves into their implications. In the independent test set, our model consistently surpasses state-of-the-art approaches across key evaluation metrics, underscoring the efficacy of its architecture. The contribution of each component within our framework is further substantiated through rigorous ablation studies, which confirms their essential role in achieving the model’s superior performance. To ensure model interpretability, UMAP is employed for high-dimensional feature visualization. The results reveal that TBC-ac4C learns a highly discriminative feature space, establishing clear decision boundaries between ac4C and non-ac4C instances.

Building on these findings, the TBC-ac4C model is inherently designed to support variable-length sequence inputs without requiring any modifications to its core modules. The Transformer encoder, TCN, and BiGRU are fundamentally sequence-to-sequence models capable of processing inputs of arbitrary length. Subsequently, a global max-pooling layer aggregates these variable-length representations into a fixed-dimension vector, providing a standardized input for the downstream classification layer. This design decouples the model’s architecture from a dependency on specific sequence lengths. Therefore, when applied to new datasets with different lengths, the TBC-ac4C framework can be adapted simply by retraining, without any need for structural adjustments, which fully demonstrates the robustness and scalability of its design.

Furthermore, a clear separation between positive and negative samples and thereby providing a visual rationale for its predictive power. This strong learning capability underpins the model’s robust generalization across diverse datasets, including both balanced and imbalanced scenarios. A critical finding from our evaluation on the imbalanced data, which mimics real-world biological conditions, is a strategic performance trade-off. While TBC-ac4C’s ACC (82.98 %) is marginally second to STM-ac4C (83.19 %), it achieves the highest Sn (89.29 %) and superior AUC (0.8962). This deliberate prioritization of sensitivity is a key features design, intended to minimize false negatives in biological discovery pipelines where the cost of a missed true site is significantly higher than that of a false positive.

5. Conclusions

This paper introduces TBC-ac4C, an integrated deep learning architecture designed for identifying ac4C sites in mRNA. Its hybrid architecture employs a Transformer encoder to capture global sequence dependencies, which are then processed by parallel TCN and BiGRU branches to model temporal and contextual patterns, respectively. A pivotal cross-attention mechanism adaptively integrates these distinct feature sets to inform the final prediction. In the benchmark dataset, TBC-ac4C model consistently outperforms state-of-the-art methods. UMAP visualization demonstrates the framework’s ability to learn highly discriminative representations, while elucidating its underlying feature separation mechanism. Furthermore, the model exhibits exceptional generalization capabilities on both balanced and imbalanced datasets.

Despite these promising results, we acknowledge avenues for future improvement. Future efforts will be directed toward: (i) Adaptive feature learning: Our ablation studies indicate a greater contribution from the TCN branch, a nuance the current static cross-attention overlooks. We will therefore introduce a dynamic gating mechanism to adaptively weight the TCN and BiGRU features. (ii) Cross-species validation: Given the training data is mainly from human cell lines, we will conduct systematic cross-species validation to rigorously evaluate the model’s applicability. (iii) Multi-modal data fusion: To overcome sequence-only limitations, we plan to integrate heterogeneous data,

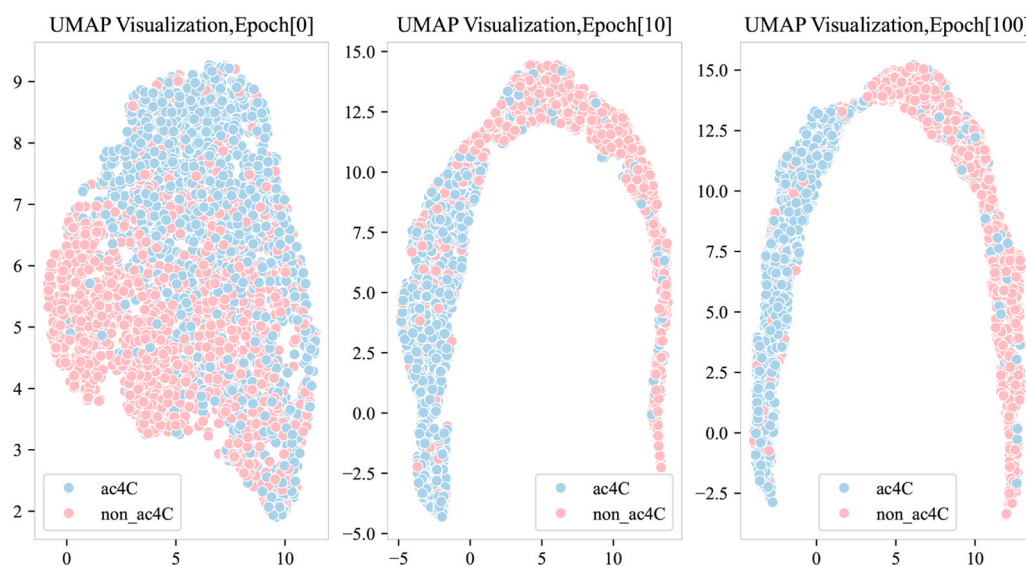


Fig. 6. Visualization of the training phase on the benchmark set using UMAP.

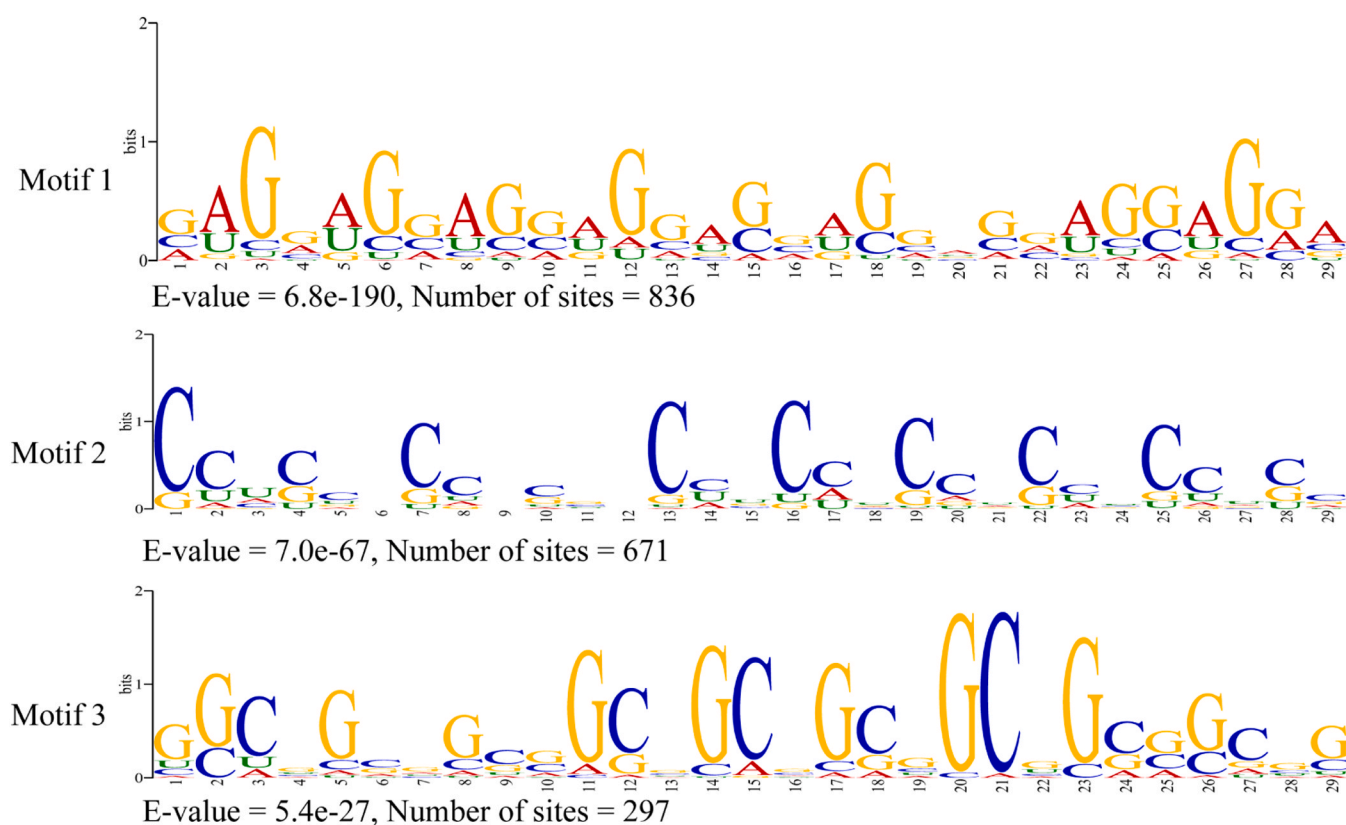


Fig. 7. Identification of conserved motifs at ac4C modification sites.

such as transcriptome expression profiles and RNA secondary structures. (iv) Advanced hyperparameter optimization: While this study employed the Optuna framework, future work will explore more advanced algorithms, such as hippopotamus optimization (HO) (Quan et al., 2026), hybrid dung beetle optimization algorithm with multiple strategies (QHDBO) (Quan et al., 2025a), and crested porcupine optimizer (CPO) (Quan et al., 2025b). These optimizers may identify a superior hyperparameter set to further enhance the model's performance and robustness.

Author contributions

Shengli Zhang conceived the study and provided overall supervision. Xinyi Zhang carried out the computational work. Both authors were involved in drafting and revising the manuscript, and endorsed the final version.

CRediT authorship contribution statement

Xinyi Zhang: Writing – review & editing, Writing – original draft,

Table 7
Comparative evaluation of multiple methods on the balanced dataset.

Method	ACC(%)	Sn(%)	Sp(%)	MCC	AUC
PACES	53.21	8.12	98.29	0.1484	0.5322
XG-ac4C	76.33	58.23	94.43	0.5650	0.7633
DeepAc4C	79.19	82.80	75.58	0.5857	0.8649
MetaAc4C-noGAN	80.94	81.80	80.09	0.6189	0.8651
MetaAc4C	81.79	79.22	84.36	0.6368	0.8749
STM-ac4C	82.01	82.44	81.58	0.6402	0.8804
TBC-ac4C	84.48	83.30	85.65	0.6897	0.9080

The bold font denotes the optimal value obtained in each column.

Table 8
Comparative evaluation of multiple methods on the unbalanced dataset.

Method	ACC(%)	Sn(%)	Sp(%)	MCC	AUC
PACES	79.26	7.28	95.49	0.0487	0.5139
XG-ac4C	80.60	12.63	95.35	0.1280	0.5399
DeepAc4C	80.51	79.65	81.37	0.6103	0.8640
MetaAc4C-noGAN	81.04	81.56	80.94	0.6209	0.8686
MetaAc4C	82.87	80.94	84.80	0.6579	0.8951
STM-ac4C	83.19	78.80	87.58	0.6664	0.8942
TBC-ac4C	82.98	89.29	76.66	0.6649	0.8962

The bold font denotes the optimal value obtained in each column.

Methodology, Formal analysis. **Shengli Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Natural Science Basic Research Program of Shaanxi (No. 2024JC-YBMS-004) and Xidian University Specially Funded Project for Interdisciplinary Exploration (No. TZJH2024028).

Data and code availability

The complete dataset and source code developed for this study are accessible via <https://github.com/xinyizhang186/TBC-ac4C>.

References

T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," 2019, arXiv:1907.10902. [Online]. Available: <https://arxiv.org/abs/1907.10902>.

Alam, W., Tayara, H., Chong, K.T., 2020. XG-ac4C: identification of N4-acetylcytidine (ac4C) in mRNA using eXtreme gradient boosting with electron-ion interaction pseudopotentials. *Sci. Rep.* 10, 20942.

Arango, D., Sturgill, D., Alhusaini, N., Dillman, A.A., Sweet, T.J., Hanson, G., et al., 2018. Acetylation of cytidine in mRNA promotes translation efficiency. *Cell* 175, 1872–1886.

Arzumanian, V.A., Dolgalev, G.V., Kurbatov, I.Y., Kiseleva, O.I., Poverennaya, E.V., 2022. Epitranscriptome: review of top 25 most-studied RNA modifications. *Int. J. Mol. Sci.* 23, 13851.

Bai, S., Koltner, J.Z., Koltun, V., 2018. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv 1803, 01271. (<https://arxiv.org/abs/1803.01271>).

Bailey, T.L., Johnson, J., Grant, C.E., Noble, W.S., 2015. The MEME suite. *Nucleic Acids Res.* 43, W39–W49.

Bhargava, P., Fitzgerald, K.C., Venkata, S.L.V., Smith, M.D., Kornberg, M.D., Mowry, E. M., et al., 2018. Dimethyl fumarate treatment induces lipid metabolism alterations that are linked to immunological changes. *Ann. Clin. Transl. Neurol.* 6, 33–45.

Boccaletto, P., Stefaniak, F., Ray, A., Cappannini, A., Mukherjee, S., Purta, E., et al., 2022. MODOMICS: a database of RNA modification pathways. *Nucleic Acids Res* 50, 231–235.

Chen, C.F., Fan, Q., Panda, R., 2021. CrossViT: cross-attention multi-scale vision transformer for image classification. arXiv 2103, 14899. (<https://arxiv.org/abs/2103.14899>).

David, R., Burgess, A., Parker, B., Li, J., Pulsford, K., Sibbritt, T., et al., 2017. Transcriptome-wide mapping of RNA 5-methylcytosine in Arabidopsis mRNAs and noncoding RNAs. *Plant Cell* 29, 445–460.

Frazier, P.I., 2018. A tutorial on Bayesian optimization. arXiv. (<https://arxiv.org/abs/1807.02811>), 1807.02811.

He, W., Han, Y., Zuo, Y., Guo, F., 2024. NBCR-ac4C: a deep learning framework based on multivariate BERT for human mRNA N4-acetylcytidine sites prediction. *J. Chem. Inf. Model* 64, 8074–8081.

Ito, S., Akamatsu, Y., Noma, A., Kimura, S., Miyauchi, K., Ikeuchi, Y., et al., 2014. A single acetylation of 18 S rRNA is essential for biogenesis of the small ribosomal subunit in *Saccharomyces cerevisiae*. *J. Bio. Chem.* 289, 26201–26212.

Jia, J., Cao, X., Wei, Z., 2023. DLC-ac4C: a prediction model for N4-acetylcytidine sites in human mRNA based on DenseNet and bidirectional LSTM methods. *Curr. Genom.* 24, 171–186.

Jia, Y., Zhang, Z., Yan, S., Zhang, Q., Wei, L., Cui, F., 2024. Voting-ac4C: pre-trained large RNA language model enhances RNA N4-acetylcytidine site prediction. *Int. J. Biol. Macromol.* 282, 136940.

Jiang, X., Cheng, Y., Zhu, Y., Xu, C., Li, Q., Xing, X., et al., 2023. Maternal NAT10 orchestrates oocyte meiotic cell-cycle progression and maturation in mice. *Nat. Commun.* 14, 3729.

Jin, G., Xu, M., Zou, M., Duan, S., 2020. The processing, gene regulation, biological functions, and clinical relevance of N4-Acetylcytidine on RNA: a systematic review. *Mol. Ther. Nucleic Acids* 20, 13–24.

Kingma, D.P., Ba, J., 2014. Adam: a method for stochastic optimization. arXiv 1412, 6980. (<https://arxiv.org/abs/1412.6980>).

Lai, F.L., Gao, F., 2023. LSA-ac4C: a hybrid neural network incorporating double-layer LSTM and self-attention mechanism for the prediction of N4-acetylcytidine sites in human mRNA. *Int. J. Biol. Macromol.* 253, 126837.

Li, W., Godzik, A., 2006. Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences. *Bioinformatics* 22, 1658–1659.

Li, C., He, Y., Li, X., Jing, X., 2019. BiGRU network for human activity recognition in high resolution range profile. 2019 International Radar Conference (RADAR), pp. 1–5.

Li, Z., Jin, B., Fang, J., 2024. MetaAc4C: a multi-module deep learning framework for accurate prediction of N4-acetylcytidine sites based on pre-trained bidirectional encoder representation and generative adversarial networks. *Genomics* 116, 110749.

Li, B., Li, D., Cai, L., Zhou, Q., Liu, C., Lin, J., et al., 2023. Transcriptome-wide profiling of RNA N4-cytidine acetylation in Arabidopsis thaliana and Oryza sativa. *Mol. Plant* 16, 1082–1098.

McInnes, L., Healy, J., Melville, J., 2018. UMAP: uniform manifold approximation and projection for dimension reduction. arXiv 1802, 03426. (<https://arxiv.org/abs/1802.03426>).

Morena, F., Argentati, C., Bazzucchi, M., Emiliani, C., Martino, S., 2018. Above the epitranscriptome: RNA modifications and stem cell identity. *Genes* 9, 329.

Pascanu, R., Mikolov, T., Bengio, Y., 2012. On the difficulty of training Recurrent Neural Networks. arXiv 1211, 5063. (<https://arxiv.org/abs/1211.5063>).

Pham, N.T., Terrance, A.T., Jeon, Y.J., Rakkiyappan, R., Manavalan, B., 2024. ac4C-AFL: a high-precision identification of human mRNA N4-acetylcytidine sites based on adaptive feature representation learning. *Mol. Ther. Nucleic Acids* 35, 102192.

PyTorch: ReduceLROnPlateau PyTorch 1.9.0 documentation, (https://pytorch.org/docs/stable/generated/torch.optim.lr_scheduler.ReduceLROnPlateau.html).

Quan, R., Cheng, G., Guan, X., Zhang, G., Quan, J., 2026. A HO-BiGRU-transformer based PEMFC degradation prediction method under different current conditions. *Renew. Energy*, 256, 124132.

Quan, R., Zhou, Y., Li, W., Wan, H., 2025a. A hybrid data-driven approach for SOH estimation of lithium-ion batteries based on health index extraction and QHDBO-BiTCN-BiGRU. *J. Energy Storage*, 125, 116788.

Quan, R., Zhou, Y., Yao, S., Wan, H., Chang, Y., 2025b. Improving the performance of a polygonal automobile exhaust thermoelectric generator with a crested porcupine optimizer. *Appl. Therm. Eng.* 268, 125946.

Roundtree, L.A., Evans, M.E., Pan, T., He, C., 2017. Dynamic RNA modifications in gene expression regulation. *Cell* 169, 1187–1200.

Stoeckle, D., Thellmann, M., Vermeer, J.E., 2018. Breakout-lateral root emergence in Arabidopsis thaliana. *Curr. Opin. Plant Biol.* 41, 67–72.

Su, W., Xie, X.Q., Liu, X.W., Gao, D., Ma, C.Y., et al., 2023. iRNA-ac4C: a novel computational method for effectively detecting N4-acetylcytidine sites in human mRNA. *Int. J. Biol. Macromol.* 227, 1174–1181.

Thomas, J.M., Briney, C.A., Nance, K.D., Lopez, J.E., Thorpe, A.L., Fox, S.D., et al., 2018. A chemical signature for cytidine acetylation in RNA. *J. Am. Chem. Soc.* 140, 12667–12670.

Thomas, G., Gordon, J., Rogg, H., 1978. N4-Acetylcytidine. a previously unidentified labile component of the small subunit of eukaryotic ribosomes. *J. Biol. Chem.* 253, 1101–1105.

A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez et al., "Attention is all you need," 2013, arXiv:1706.03762. [Online]. Available: <https://arxiv.org/abs/1706.03762>.

Wang, C., Ju, Y., Zou, Q., Lin, C., 2021. DeepAc4C: a convolutional neural network model with hybrid features composed of physicochemical patterns and distributed representation information for identification of N4-acetylcytidine in mRNA. *Bioinformatics* 38, 52–57.

- Xie, L., Zhong, X., Cao, W., Liu, J., Zu, X., Chen, L., 2023. Mechanisms of NAT10 as ac4C writer in diseases. *Mol. Ther. Nucleic Acids* 32, 359–368.
- Yi, M., Zhou, F., Deng, Y., 2024. STM-ac4C: a hybrid model for identification of N4-acetylcytidine (ac4C) in human mRNA based on selective kernel convolution, temporal convolutional network, and multi-head self-attention. *Front. Genet.* 15, 1408688.
- Yuan, J., Wang, Z., Pan, Z., Li, A., Zhang, Z., Cui, F., 2024. DPNN-ac4C: a dual-path neural network with self-attention mechanism for identification of N4-acetylcytidine (ac4C) in mRNA. *Bioinformatics* 40, btae625.
- Zhang, S., Liu, K., Xu, Y., 2025. TransCNN: a novel architecture combining transformer and TextCNN for detecting N4-acetylcytidine sites in human mRNA. *Anal. Biochem.* 703, 115882.
- Zhang, X., Zeng, J., Wang, J., Yang, Z., Gao, S., Liu, H., et al., 2022. Revealing the potential markers of N(4)-acetylcytidine through acRIP-seq in triple-negative breast cancer. *Genes* 13, 2400.
- Zhao, W., Zhou, Y., Cui, Q., Zhou, Y., 2019. PACES: prediction of N4-acetylcytidine (ac4C) modification sites in mRNA. *Sci. Rep.* 9, 11112.